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INFLUENCE OF ELECTRON-PHONON INTERACTION ON THE MOTION OF AN ELECTRON IN A ONE-DIMENSIONAL MOLECULAR SYSTEM

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The influence of the electron-phonon interaction on the motion of an electron in a one-dimensional molecular chain is investigated. It is shown that in the case of strong interaction the motion of the electron with velocity less than the velocity of a longitudinal acoustic wave in the chain is accompanied by the motion of a local deformation and has a quasisuperconducting nature. In molecular chains with large anharmonicity, it is possible to have the formation of acoustic solitons, which are displaced with velocity exceeding the velocity of sound. Such solitons can participate in the electron transport. It is suggested that the investigated phenomena could play a part in biological processes in the transport of molecules through protein molecules from donor to acceptor molecules.

1. Introduction

Many biological processes (photosynthesis, tissue breathing, the action of enzymes, among others) involve the transport of electrons from donors to acceptors along biological structures, which are called electron transport chains.

Generally it is assumed that this electron transport is realized by the mechanism of quantum-mechanical tunneling. Such an idea was first suggested in [1]. A detailed theoretical investigation of the possibility of tunnel transport of an electron between two links in an electron transport chain was made in the adiabatic approximation by Grigorov and Chernavskii [2].

In a number of experimental studies (see, for example, [3]) it has been established that the electron transfer is realized over distances 30-70 Å. The centers between which the electron is transferred are separated by many atoms and atomic groups, which are coupled to each other by covalent, hydrogen or van der Waals forces. One of the possible explanations of electron transport over large distances involves the assumption that the process is facilitated by protein molecules between the donor and acceptor electrons.

The donor-acceptor model of electron transport through protein molecules has been most fully developed by Petrov, Ukrainskii, and Kharkyanen [4]. Besides below-barrier electron transport, the process of thermal transfer of an electron to the conduction band with subsequent transition to an acceptor has been investigated. In [5], the present author investigated below-barrier transport of an electron from a donor to an acceptor with the participation of a protein molecule. A model of a quasioone-dimensional molecular system with one extra electron was used in these studies as a protein molecule.

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The motion of an extra electron in quasione-dimensional molecular systems is usually investigated in the adiabatic approximation, i.e., for a fixed periodic arrangement of the molecules. In this case, the interaction of the electron with the displacements of the molecules from the equilibrium positions is treated as a perturbation that slows down the motion of the electron. In the present paper, we investigate the motion of an electron along a quasione-dimensional molecular chain without using perturbation theory. This has the consequence that the motion of the electron is described by means of nonlinear equations. Their solution shows that the motion of the electron is accompanied by local deformation of the chain. It is described by a wave function in the form of a soliton.

As a result of the coupling of the electron to the local deformation of the chain, the electron mass increases, and the energy is lowered relative to the bottom of the conduction band. The electron moves with a velocity less than the velocity of longitudinal sound, and does not lose energy on the emission of phonons. This stabilization of the electron motion is particularly pronounced in soft chains in the case of strong coupling between the electron and the deformation of the chain. In these cases, the motion of the electron recalls in a certain sense the motion of electrons in the superconducting state of a metal, which is also due to interaction of an electron with a deformation of the crystal lattice.

In molecular chains with anharmonic interaction between the molecules and dispersion of the acoustic waves, excitations of the type of acoustic solitons are possible. Such solitons are local deformations of the chain that are displaced with a velocity exceeding the velocity of longitudinal sound in the chain. In some cases, such solitons may capture an electron and transport it along the chain with a velocity exceeding the velocity of sound. This phenomenon is considered in the fourth and fifth sections of this paper.

2. Basic Equations Determining the Motion of an Additional Electron in a Quasione-Dimensional Molecular Chain

We denote the equilibrium positions in the infinite chain of identical molecules by the vectors $\mathbf{n} = \mathbf{an}$, where $n = 0, \pm 1, \pm 2, \dots$. We use the model of strongly coupled electrons, in which the state of the additional electron in the chain of molecules occupying the sites $\mathbf{n}$ is characterized by the Hamiltonian

$$H_{el} = \sum_n [(\epsilon_0 - D_0) b_n^+ b_n - D_a (b_{n+a}^+ b_n + b_n^+ b_{n+a})], \quad (2.1)$$

where b_n^+, b_n are the operators of the presence and absence of the additional electron at site $\mathbf{n}$; ϵ_0 and $b_n^+ |0\rangle = \varphi(\mathbf{r} - \mathbf{n})$ are the energy and wave function of this electron in an isolated molecule.

If $W(\mathbf{r} - \mathbf{n})$ is the operator of the potential interaction energy of an electron having coordinate $\mathbf{r} - \mathbf{n}$ with the neighboring site $\mathbf{n} + \mathbf{a}$, then*

$$D_0 = - \int |\varphi(\mathbf{r} - \mathbf{n})|^2 [W(\mathbf{r} - \mathbf{n} + \mathbf{a}) + W(\mathbf{r} - \mathbf{n} - \mathbf{a})] d^3r \geq 0 \quad (2.2)$$

characterizes the energy of the interaction of an electron near site $\mathbf{n}$ with the neighboring molecules of the chain; the integral

$$D_a = - \int \varphi^*(\mathbf{r} - \mathbf{n} - \mathbf{a}) W(\mathbf{r} - \mathbf{n} - \mathbf{a}) \varphi(\mathbf{r} - \mathbf{n}) d^3r \quad (2.3)$$

characterizes the energy of the exchange transport of the electron between neighboring molecules. Below, we consider the case when $D_a > 0$.

Let u_n be small longitudinal displacements of the molecules from the equilibrium positions $\mathbf{n}$; then the operator of the energy of the displacements of the molecules in the harmonic approximation has the form

$$H_{ph} = \frac{1}{2} \sum_n \left[\frac{1}{M} \hat{p}_n^2 + \kappa_2 (u_n - u_{n-a})^2 \right], \quad (2.4)$$

where M is the mass of the molecule, κ_2 is the coefficient of longitudinal elasticity, and $\hat{p}_n$ is the momentum operator conjugate to the displacement operator u_n . They satisfy the commutation relations $[u_n, \hat{p}_{n'}] = i\hbar \delta_{nn'}$.

* Here and in all the following expressions, the integration is within infinite limits.

On the displacement of molecules from the equilibrium positions, an interaction in addition to (2.2) and (2.3) arises between the electron and the molecules. The operator of this interaction can be written in the simplest case in the form

$$H_{\text{int}} = \chi \sum_{\mathbf{n}} b_{\mathbf{n}}^+ b_{\mathbf{n}} (u_{\mathbf{n}+\mathbf{a}} - u_{\mathbf{n}-\mathbf{a}}). \quad (2.5)$$

Thus, the states of our system – the electron and the chain of molecules – are described by the Hamiltonian

$$H = H_{\text{el}} + H_{\text{ph}} + H_{\text{int}}. \quad (2.6)$$

We shall seek the wave functions characterizing the stationary states of the system in the form

$$\psi(t) = \sum_{\mathbf{n}} A_{\mathbf{n}}(t) e^{i\sigma(t)} b_{\mathbf{n}}^+ |0\rangle, \quad \sum_{\mathbf{n}} |A_{\mathbf{n}}(t)|^2 = 1. \quad (2.7)$$

In this expression, $|0\rangle$ is the vacuum state for the phonons and the electrons; $\sigma(t)$ is an operator of the form

$$\sigma(t) = -\frac{i}{\hbar} \sum_{\mathbf{n}} [\beta_{\mathbf{n}}(t) \hat{p}_{\mathbf{n}} - \pi_{\mathbf{n}}(t) u_{\mathbf{n}}]. \quad (2.8)$$

The squares of the moduli of the functions $A_{\mathbf{n}}(t)$ characterize the probabilities of finding an electron at site $\mathbf{n}$. The functions $\beta_{\mathbf{n}}(t)$ and $\pi_{\mathbf{n}}(t)$ determine the mean equilibrium values in the state (2.7) of the momenta and the displacements of the molecules from the equilibrium positions. Indeed, it is easy to show that $\beta_{\mathbf{n}}(t) = \langle \psi(t) | u_{\mathbf{n}} | \psi(t) \rangle$, $\pi_{\mathbf{n}}(t) = \langle \psi(t) | \hat{p}_{\mathbf{n}} | \psi(t) \rangle$. Using in what follows the quantities $\beta_{\mathbf{n}}(t)$ and $\pi_{\mathbf{n}}(t)$, we naturally do not take into account the zero-point vibrations of the molecules with respect to the equilibrium positions.

To find the unknown functions $A_{\mathbf{n}}(t)$, $\beta_{\mathbf{n}}(t)$, and $\pi_{\mathbf{n}}(t)$ we consider the functional

$$\Phi_2 = \langle \psi | H | \psi \rangle = \sum_{\mathbf{n}} \left\{ A_{\mathbf{n}}^* [(\varepsilon_{\mathbf{n}} - D_0) A_{\mathbf{n}} - D_{\mathbf{n}} (A_{\mathbf{n}+\mathbf{a}} + A_{\mathbf{n}-\mathbf{a}})] + \chi |A_{\mathbf{n}}|^2 (\beta_{\mathbf{n}+\mathbf{a}} - \beta_{\mathbf{n}-\mathbf{a}}) + \frac{1}{2} \left(\frac{1}{M} \pi_{\mathbf{n}}^2 + \kappa_2 \rho_{\mathbf{n}}^2 \right) \right\}, \quad (2.9)$$

where

$$\rho_{\mathbf{n}} = \beta_{\mathbf{n}-\mathbf{a}} - \beta_{\mathbf{n}}. \quad (2.10)$$

The expression (2.9) is the Hamilton function of the variables $A_{\mathbf{n}}$ and $\beta_{\mathbf{n}}$ and their canonically conjugate generalized momenta $i\hbar A_{\mathbf{n}}^*$, $\pi_{\mathbf{n}}$. Therefore, using (2.9), we find the Hamilton equations

$$i\hbar \partial A_{\mathbf{n}} / \partial t = [\varepsilon_{\mathbf{n}} - D_0 - \chi (\beta_{\mathbf{n}+\mathbf{a}} - \beta_{\mathbf{n}-\mathbf{a}})] A_{\mathbf{n}} - D_{\mathbf{n}} (A_{\mathbf{n}+\mathbf{a}} - A_{\mathbf{n}-\mathbf{a}}), \quad (2.11)$$

$$\partial \beta_{\mathbf{n}} / \partial t = \pi_{\mathbf{n}} / M, \quad (2.11a)$$

$$\partial \pi_{\mathbf{n}} / \partial t = -\chi (|A_{\mathbf{n}-\mathbf{a}}|^2 - |A_{\mathbf{n}+\mathbf{a}}|^2) - \kappa_2 (\rho_{\mathbf{n}+\mathbf{a}} - \rho_{\mathbf{n}}). \quad (2.11b)$$

Differentiating (2.11a) with respect to the time and using (2.11b), we find

$$\partial^2 \beta_{\mathbf{n}} / \partial t^2 + \chi (|A_{\mathbf{n}-\mathbf{a}}|^2 - |A_{\mathbf{n}+\mathbf{a}}|^2) + \kappa_2 (\rho_{\mathbf{n}+\mathbf{a}} - \rho_{\mathbf{n}}) = 0. \quad (2.12)$$

The system of equations (2.11) and (2.12) determines the unknown functions $A_{\mathbf{n}}(t)$, $\beta_{\mathbf{n}}(t)$, and $\pi_{\mathbf{n}}(t)$, in terms of which the wave functions (2.7) of the stationary states of the chain of molecules with one external electron are determined. Substituting the obtained values of $A_{\mathbf{n}}$, $\beta_{\mathbf{n}}$, and $\pi_{\mathbf{n}}$ in (2.9), we find the energy of the system measured from the energy of the vacuum state.

3. Continuum Approximation*

We consider solutions of the system of equations (2.11) and (2.12) that vary smoothly over a distance a . In this case, we can replace the functions $A_{\mathbf{n}}(t)$ and $\beta_{\mathbf{n}}(t)$ of the discrete variable $\mathbf{n}$ by functions of a continuous variable x measured along the direction $\mathbf{a}$. For $x = na$, the equations $A_{\mathbf{n}}(t) = A(x, t)$, $\beta_{\mathbf{n}}(t) = \beta(x, t)$ must be satisfied. According to (2.10), the function characterizing the decrease in the intermolecular distances in the chain is now determined by the expression

$$\rho(x, t) = -a \partial \beta(x, t) / \partial x. \quad (3.1)$$

* The motion of an external electron in a one-dimensional dielectric crystal in the continuum approximation has also been considered by Kosevich [6].

In the continuum approximation, Eqs. (2.11) and (2.12) take the following form on the transition by means of (3.1) from the variables $\beta(x, t)$ to the variables $\rho(x, t)$:

$$\left[i\hbar \frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m^*} \frac{\partial^2}{\partial x^2} + 2\chi\rho \right] A(x, t) = 0, \quad (3.2)$$

$$\left(\frac{\partial^2}{\partial t^2} - V_a^2 \frac{\partial^2}{\partial x^2} \right) \rho(x, t) + \frac{2\chi a^2}{M} \frac{\partial^2}{\partial x^2} |A(x, t)|^2 = 0. \quad (3.3)$$

In writing down these equations, we have introduced the abbreviated notation

$$\mathcal{E}_0 = \varepsilon_a - D_0 - 2D_a, \quad (3.4)$$

$$\hbar^2/2m^* = a^2 D_a, \quad (3.5)$$

$$V_a^2 = a^2 \kappa_2 / M. \quad (3.6)$$

Suppose that $\chi = 0$, i.e., there is no interaction of the electron with the displacements of the molecules from the equilibrium positions. In this case, Eqs. (2.2) and (2.3) become independent. In particular, the equation $(\partial^2/\partial t^2 - V_a^2 \partial^2/\partial x^2) \rho(x, t) = 0$ describes longitudinal acoustic waves $\rho(x, t) = \rho_0 \sin(t \mp x/V_a)$ propagating with velocity V_a in the positive or negative direction of the x axis. Equation (3.2) for $\chi = 0$ has solutions in the form of plane waves $A_k(x, t) = A_0 \exp\{i[kx - \omega(k)t]\}$, corresponding to an energy of the electron near the bottom of the conduction band, $\hbar\omega(k) = \mathcal{E}_0 + \hbar^2 k^2/2m^*$, $ka \ll 1$, with effective mass m^* and momentum $\hbar k$. The value $\mathcal{E}_0$ corresponds to the energy of the bottom of the conduction band.

We now consider the coupled system of equations (3.2) and (3.3). By virtue of the translational invariance of the chain, we can seek solutions of this system in the form of excitations propagating with constant velocity V . Setting $\rho(x, t) = \rho(x - Vt)$, $|A(x, t)|^2 = f(x - Vt)$ and bearing in mind that $\rho(\xi)$ is zero for all values of ξ for which $f(\xi) = 0$, we find from Eq. (3.3), after double integration with respect to the variable $\xi = x - Vt$,

$$\rho(\xi) = \frac{2\chi}{\kappa_2(1-s^2)} |A|^2, \quad (3.7)$$

where $s = V/V_a$. Substituting (3.7) in Eq. (3.2), we obtain the nonlinear Schrödinger equation

$$\left\{ i\hbar \frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m^*} \frac{\partial^2}{\partial x^2} + G(s) |A(x, t)|^2 \right\} A(x, t) = 0 \quad (3.8)$$

with nonlinearity parameter

$$G(s) = G(0)/(1-s^2), \quad (3.9)$$

where $G(0) = 4\chi^2/\kappa_2$.

For positive value of $G(s)$, i.e., $s < 1$, the term $G(s) |A(x, t)|^2$ in Eq. (3.8) plays the part of a negative potential and leads to localization of the excitation. If $s > 1$, then $G(s) < 0$, and therefore solutions normalized by the condition

$$\frac{1}{a} \int |A(x, t)|^2 dx = 1 \quad (3.10)$$

are absent. The non-normalized solution in the form of the plane wave $A(x, t) = A_0 \exp\{i(kx - \omega t)\}$ has in this case the large energy $\hbar\omega = \mathcal{E}_0 - |A_0|^2 G(s) + \hbar k^2/2m^*$, $G(s) < 0$.

The solution of Eq. (3.8) for $s < 1$ normalized by the condition (3.10) has the form

$$A(x, t) = \frac{\sqrt{\mu a} \exp\{i[k\xi - (\omega - kV)t]\}}{\sqrt{2 \operatorname{ch} \mu \xi}}, \quad (3.11)$$

where

$$k = m^*V/\hbar, \quad \xi = x - x_0 - Vt, \quad (3.12)$$

$$\mu = am^*G(s)/2\hbar^2 = \chi^2/a\kappa_2(1-s^2)D_a, \quad (3.13)$$

$$\hbar\omega = \mathcal{E}_0 + \hbar^2(k^2 - \mu^2)/2m^*. \quad (3.14)$$

Here, $\Delta x = \pi/\mu$ characterizes the region of localization of the electron. Our continuum approximation is determined by fulfillment of the inequality

$$\Delta x \approx \pi \kappa_2 a D_a (1-s^2)/\chi^2 \gg a. \quad (3.15)$$

For fixed parameters of the chain (κ_2 , χ , D_a), this inequality imposes a restriction on the displacement velocity of the electron.

In accordance with (3.7) and (3.11), the decrease of the intermolecular distances is determined by the expression

$$\rho(\xi) = a\mu\chi/(1-s^2)\kappa_2 \operatorname{ch}^2 \mu\xi. \quad (3.16)$$

The displacements of the molecules from their equilibrium positions is represented by the function

$$\beta(\xi) = \frac{\chi}{\kappa_2(1-s^2)}[1 - \operatorname{th} \mu\xi]. \quad (3.16a)$$

From (3.1) and (3.12) we have the useful equation

$$\partial\beta(\xi)/\partial t = V\rho(\xi)/a. \quad (3.17)$$

Using (3.12), we can transform the functional (2.9) in the continuum approximation to the form

$$\Phi_2 = \frac{1}{a} \int \left\{ A^* \left[\mathcal{E}_0 - a^2 D_a \frac{\partial^2}{\partial x^2} \right] A - 2\chi |A|^2 \rho + \frac{\kappa_2(1+s^2)}{2} \rho^2 \right\} d\xi. \quad (3.18)$$

Substituting in (3.18) the values (3.11) and (3.16), we obtain the energy of the system:

$$E(V) = \mathcal{E}_0 + \frac{1}{2} m^* V^2 - 2\chi^4 (1-3s^2)/3\kappa_2^2 (1-s^2) D_a. \quad (3.19)$$

For $s \ll 1$, this expression reduces to

$$E(V) = \mathcal{E}_0 - 2\chi^4/3\kappa_2^2 D_a + \frac{1}{2} m_{sol} V^2, \quad (3.20)$$

where

$$m_{sol} = m^* + 4M\chi^4/3\kappa_2^2 D_a a^2. \quad (3.21)$$

Thus, in the state (3.11) the motion of the electron is accompanied by a local deformation of the molecular chain. In accordance with (3.21), the effective mass m^* of the electron is increased. For $V = 0$, the total energy of the electron and the chain is reduced because of their interaction by the amount $2\chi^4/3\kappa_2^2 D_a$. Moving with velocity V less than the velocity of longitudinal sound, the electron does not emit phonons. Thus, the interaction of the electron with the deformation of the chain stabilizes its motion (quasisuperconducting state). For small values of κ_2 and large value of the parameter χ , this stabilization is very important.

4. Motion of an Electron in a Chain with Cubic Anharmonicity

We consider the case when an anharmonic term of the following form (cubic anharmonicity) is added to the harmonic operator (2.4) of the energy of the displacements of the molecules from their equilibrium positions:

$$- \frac{1}{3} \kappa_3 \sum_n \rho_n^3 \quad \rho_n = \beta_{n+a} - \beta_n. \quad (4.1)$$

In this case, the functional (2.9) is replaced by the functional

$$\Phi_3 = \Phi_2 - \frac{1}{3} \kappa_3 \sum_n \rho_n^3, \quad \kappa_3 \ll \kappa_2. \quad (4.2)$$

The variation in time of the functions $A_n(t)$ is determined in this case too by Eq. (2.11). Equation (2.12), which characterizes the change in the mean values of the displacements β_n , is now replaced by the equation

$$M \partial^2 \beta_n / \partial t^2 + \chi (|A_{n+a}|^2 - |A_{n-a}|^2) + \kappa_2 (2\beta_n - \beta_{n+a} - \beta_{n-a}) + \kappa_3 (\rho_{n+a}^2 - \rho_{n-a}^2) = 0. \quad (4.3)$$

Equation (4.3) contains small anharmonic terms, and therefore, on the transition to the continuum approximation, we must retain in the expansions of the functions $\beta(x \mp a)$ the terms including the fourth derivative. Then Eq. (4.3) is transformed to

$$M \frac{\partial^2 \beta}{\partial t^2} - a^2 \kappa_2 \frac{\partial^2 \beta}{\partial x^2} - \frac{a^4 \kappa_2}{12} \frac{\partial^4 \beta}{\partial x^4} + a \kappa_3 \frac{\partial}{\partial x} \rho^2 = 2\chi a \frac{\partial}{\partial x} |A|^2.$$

Applying the operator $-a\partial/\partial x$ to both sides of this equation and using (3.1) and the definition (3.6), we obtain the equation for the functions $\rho(x, t)$:

$$\frac{\partial^2 \rho}{\partial t^2} - V_a^2 \left[\frac{\partial^2 \rho}{\partial x^2} + \frac{a^2}{12} \frac{\partial^4 \rho}{\partial x^4} + \frac{\kappa_3}{\kappa_2} \frac{\partial^2}{\partial x^2} \rho^2 \right] = - \frac{2\chi a^2}{M} \frac{\partial^2}{\partial x^2} |A|^2. \quad (4.4)$$

In a free harmonic chain, allowance for the term $a^2 \partial^4 \rho / \partial x^4$ is equivalent to allowance for the dispersion of the acoustic waves. Indeed, for $\kappa_3 = \chi = 0$ Eq. (4.4) is transformed into

$$\frac{\partial^2 \rho}{\partial t^2} - V_a^2 \left[\frac{\partial^2}{\partial x^2} + \frac{a^2}{12} \frac{\partial^4}{\partial x^4} \right] \rho = 0.$$

This equation has periodic solutions $\rho = \rho_0 \exp \{i[kx - \omega t]\}$ for the value $\omega^2 = k^2 V_a^2 (1 - a^2 k^2 / 12)$, which corresponds to the dispersion $\omega^2 = \frac{\kappa_2}{M} \sin^2 \frac{ka}{2}$ for $ka \ll 1$. The group velocity of the displacement of such excitations is $V = d\omega/dk = V_a (1 - a^2 k^2 / 8)$.

If an electron moves with the deformation with constant velocity V , then the functions $\rho(x, t)$ and $|A(x, t)|^2$ depend only on the variable (3.12). In this case, Eq. (4.4) can be written in the form

$$\frac{\partial^2}{\partial \xi^2} \left[\frac{a^2 \kappa_2}{12} \frac{\partial^2 \rho}{\partial \xi^2} - (s^2 - 1) \kappa_2 \rho + \kappa_3 \rho^2 - 2\chi |A|^2 \right] = 0.$$

Integrating this equation twice with respect to the variable ξ and noting that for all sufficiently large values of ξ , when $|A(\xi)|^2$ is equal to zero, there is no deformation of the chain, we obtain the equation

$$\frac{a^2 \kappa_2}{12} \frac{\partial^2 \rho}{\partial \xi^2} - (s^2 - 1) \kappa_2 \rho + \kappa_3 \rho^2 - 2\chi |A|^2 = 0. \quad (4.5)$$

We assume first that the anharmonicity is more important than the dispersion of the acoustic waves. In this case, Eq. (4.5) reduces to the quadratic equation

$$\kappa_3 \rho^2 - (s^2 - 1) \kappa_2 \rho - 2\chi |A|^2 = 0. \quad (4.6)$$

Solving this equation, we find two roots:

$$\rho_{1,2} = \frac{\kappa_2 (s^2 - 1)}{2\kappa_3} \left[1 \mp \sqrt{1 + \frac{8\chi \kappa_3 |A|^2}{\kappa_2^2 (1 - s^2)^2}} \right]. \quad (4.7)$$

At low displacement velocities of the electron, when $s \ll 1$, one can have fulfillment of the inequality $8\chi \kappa_3 |A|^2 \ll \kappa_2^2 (1 - s^2)^2$. In this case, the two roots (4.7) have the approximate values

$$\rho_1 \approx \frac{2\chi |A|^2}{\kappa_2 (1 - s^2)}, \quad \rho_2 \approx - \frac{\kappa_2 (1 - s^2)}{\kappa_3} - \frac{2\chi |A|^2}{\kappa_2 (1 - s^2)}.$$

When ρ_1 is substituted in Eq. (3.2), we obtain the already investigated nonlinear Schrödinger equation (3.8), which characterizes the slow motion of the electron accompanied by local deformation of the molecular chain. Substituting ρ_2 in Eq. (3.2), we obtain an equation that has only nonlocalized solutions characterizing states with higher excitation energy.

When allowance is made for anharmonicity and dispersion of the acoustic waves, Eq. (4.5) reduces to an equation that is usually called the Boussinesq equation,

$$\frac{a^2}{12} \frac{\partial^2 \rho_0}{\partial \xi^2} - (s^2 - 1) \rho_0 + \frac{\kappa_3}{\kappa_2} \rho_0^2 = 0. \quad (4.8)$$

Such an equation describes intrinsic excitations in a nonlinear chain without external electron. The solution of this equation was considered in [7] by Toda and Waddati. By direct substitution one can show that a solution in the form of a soliton corresponds to the minimal energy of an excitation propagating along the chain with constant velocity V :

$$\rho_0 = a^2 \mu^2 \kappa_2 / 2\kappa_3 \operatorname{ch}^2 \mu \xi \quad (4.9)$$

for

$$\xi = x - x_0 - Vt, \quad (4.10)$$

$$\mu^2 = 3(s^2 - 1)/a^2 > 0. \quad (4.11)$$

Thus, the propagation velocity of the excitation (4.9), localized in the neighborhood of $\xi \approx 0$, exceeds the velocity V_a of longitudinal sound in the harmonic chain, since $V = sV_a = sa\sqrt{\kappa_2/M}$.

The localized excitation described by the function (4.9), which characterizes a decrease in the distances between neighboring molecules in the chain, may be called an acoustic soliton. Our continuum

approximation is justified if we have fulfillment of the inequality $\pi \gg \mu a$, which restricts the admissible velocities of the acoustic solitons by the condition $1 + 1/3\pi^2 > s^2 > 1$.

The energy and momentum of a soliton that is displaced in the chain with the velocity V are determined by the expressions

$$W_s = \frac{1}{2a} \int \left[\kappa_2 \rho_0^2 + M \left(\frac{\partial \beta_0}{\partial t} \right)^2 - \frac{2\kappa_3}{\kappa_2} \rho_0^3 \right] d\xi, \quad \mathcal{P}_s = \frac{M}{a} \int \frac{\partial \beta_0}{\partial t} d\xi.$$

Substituting the expressions (3.17), (4.9), and (4.11), we obtain, respectively,

$$W_s = \frac{\sqrt{3}\kappa_2^3}{10\kappa_3^2} (s^2 - 1)^{3/2} (11s^2 + 7), \quad \mathcal{P}_s = \frac{\kappa_2 M V}{a\kappa_3} \sqrt{3(s^2 - 1)}.$$

Our solutions of Eq. (4.8) hold only for chains whose length is appreciably greater than the region (π/μ) covered by the excitation. To investigate the conditions of excitation of acoustic solitons, we must consider the influence of boundary effects, since such solitons can be excited only from the ends of the chain.

The local deformation corresponding to the acoustic soliton creates a potential well for the electron. This well can trap the external electron and displace it along the chain with a velocity somewhat exceeding the velocity of longitudinal sound. To elucidate this possibility, we substitute the value (4.9) in Eq. (3.2). We obtain the Schrödinger equation

$$\left[i\hbar \frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m^*} \frac{\partial^2}{\partial x^2} + \frac{U_0}{\text{ch}^2 \mu \xi} \right] A(x, t) = 0, \quad (4.12)$$

where

$$U_0 = 3\chi\kappa_2(s^2 - 1)/\kappa_3 > 0, \quad s > 1. \quad (4.13)$$

We introduce further the wave function $\varphi(\xi)$ by means of the relation

$$A(x, t) = \varphi(\xi) \exp\{i(kx - \omega t)\}, \quad (4.14)$$

where

$$k = m^*V/\hbar, \quad \xi = x - x_0 - Vt, \quad (4.15)$$

and $\hbar\omega$ is the electron energy in the fixed coordinate system. Substituting (4.14) in Eq. (4.12), we find a Schrödinger equation for the function $\varphi(\xi)$, which determines the motion of the electron in a coordinate system moving with the acoustic soliton,

$$\frac{\partial^2 \varphi}{\partial \xi^2} + \frac{2m^*}{\hbar^2} \left(E - \frac{U_0}{\text{ch}^2 \mu \xi} \right) \varphi = 0, \quad (4.16)$$

where

$$E = \hbar\omega - \mathcal{E}_0 - 1/2 m^* V^2 \quad (4.16a)$$

is the energy of the electron in the potential well $U_0 \text{sech}^2 \mu \xi$.

The solution of Eq. (4.16) is considered in Landau and Lifshits's book [8] (§23). The discrete spectrum of negative energies is determined by the formula

$$E_n = -\frac{\hbar^2 \mu^2}{2m^*} (v - n)^2, \quad (4.17)$$

where

$$n = 0, 1, \dots, n_{\max}, \quad n_{\max} \leq v = 1/2 (\sqrt{1 + 8a^2 \chi \kappa_2 m^* / \kappa_3 \hbar^2} - 1). \quad (4.18)$$

The lowest energy level

$$E_0 = -\hbar^2 \mu^2 v^2 / 2m^* = -3(s^2 - 1)v^2 \hbar^2 / 2m^* a^2 \quad (4.19)$$

is characterized by the wave function

$$\varphi_0(\xi) = \sqrt{N} / \text{ch}^v(\mu \xi), \quad (4.20)$$

where

$$N = a\mu \Gamma(v + 1/2) / \sqrt{\pi} \Gamma(v) \quad (4.20a)$$

is a normalization factor, and $\Gamma(x)$ is the gamma function. For $a \approx 5 \cdot 10^{-8}$ cm, the energy of the lowest electron state takes the value $E_0 \approx 1.5 \frac{m}{m^*} (s^2 - 1)v^2$ eV, where m is the mass of the free electron. Capture of an electron by an acoustic soliton is possible if $|E_0|$ exceeds the mean energy of the thermal vibrations.

We have considered the capture of an electron by an acoustic soliton without allowance for the back reaction of the electron on the soliton. To investigate this influence, we seek a solution of (4.5) in the form

$$\rho(\xi) = \rho_0(\xi) + \rho_1(\xi), \quad \rho_1 \ll \rho_0. \quad (4.21)$$

Substituting (4.21) in Eq. (4.5), ignoring the square of ρ_1 , and using (4.9) and (4.20), we obtain the linearized equation

$$\frac{a^2}{12} \frac{\partial^2 \rho_1}{\partial \xi^2} - (s^2 - 1) \rho_1 + \frac{3(s^2 - 1)}{ch^2 \mu \xi} \rho_1 = \frac{2\chi N}{\kappa_2 ch^{2\nu}(\mu \xi)}.$$

The solution of this equation has the form

$$\rho_1(\xi) = \Gamma(\nu + 1/2) 2\chi ch^2 \mu \xi / \Gamma(\nu) \sqrt{3\pi(s^2 - 1)} \kappa_2 ch^{2\nu}(\mu \xi). \quad (4.22)$$

If $\nu \geq 2$, and

$$4\chi \kappa_3 \Gamma(\nu + 1/2) / \sqrt{\pi} \Gamma(\nu) [3(s^2 - 1)]^{3/2} \kappa_2^2 \ll 1, \quad (4.23)$$

then the inequality $\rho_0 \ll \rho_1$ is satisfied for all values of ξ . In this case, to investigate the motion of the electron, we can in the first approximation use the equation of the zeroth approximation (4.8), which, as we have seen, gives a solution in the form of a soliton capable of capturing an electron and moving with velocity exceeding the velocity of sound.*

The above effect of displacement of an electron by an acoustic soliton can also play a part in biological systems when the interaction of the donor molecule with the protein molecule through which the electron is transported to the acceptor molecule is accompanied by the production of an acoustic soliton in the protein molecule. Naturally, such transport requires expenditure of energy on the formation of the soliton.

5. Motion of an Electron in a Chain with Anharmonicity of Fourth Degree

When there is anharmonicity in the chain of fourth degree, we must add to the operator (2.4) of the energy of the displacements of the molecules from their equilibrium positions a term of the form $-1/\kappa_4 \sum_n \rho_n^4$.

In this case, when allowance is made for dispersion of the acoustic waves in the continuum approximation the equation characterizing the change in the equilibrium distances $\rho(\xi)$ between the molecules has the form

$$\frac{a^2}{12} \frac{\partial^2 \rho}{\partial \xi^2} - (s^2 - 1) \rho + \frac{\kappa_4}{\kappa_2} \rho^3 = \frac{2\chi}{\kappa_2} |A|^2, \quad (5.1)$$

where $\xi = x - x_0 - Vt$.

We consider first intrinsic excitations propagating with constant velocity $V = sV_a$ along the chain. Such excitations are described by the equation

$$\frac{a^2}{12} \frac{\partial^2 \rho_0}{\partial \xi^2} - (s^2 - 1) \rho_0 + \frac{\kappa_4}{\kappa_2} \rho_0^3 = 0. \quad (5.2)$$

A solution of this equation that decreases sufficiently rapidly as $|\xi| \rightarrow \infty$ exists only for $s > 1$. For fixed $s > 1$, the solution of Eq. (5.2) with the lowest energy is doubly degenerate:

$$\rho_0(\xi) = \pm \sqrt{2(s^2 - 1) \kappa_2 \kappa_4 / ch \mu \xi}, \quad (5.3)$$

where

$$\mu^2 = 12(s^2 - 1)/a^2. \quad (5.4)$$

Our continuum approximation is justified if

$$\pi \gg a\mu = \sqrt{12(s^2 - 1)}, \quad (5.5)$$

which restricts the allowed velocities $V = sV_a$.

Excitations corresponding to positive sign in (5.3) are called solitons. When solitons are formed

* In [9], in a qualitative investigation of autolocalized states in a one-dimensional system, Ktitorov and Siparov conclude that autolocalization leading to a soliton-like solution is realized only at velocities less than the velocity of sound. Unfortunately, they do not indicate what values of the parameters of the system they used in the calculations.

in the excitation region, a contraction occurs, i.e., the distances between the molecules decrease. Excitations corresponding to a negative sign in (5.3) are called antisolitons. When antisolitons are formed in the excitation region, expansion occurs.

When allowance is made for Eq. (3.17), the energy transported by the solitons and antisolitons is determined by the expression

$$W_s = \frac{1}{2a} \int \left[\kappa_2(1+s^2)\rho_0^2 - \frac{\kappa_4}{2}\rho_0^4 \right] d\xi.$$

Substituting the value (5.3), we find $W_s = (5+s^2)\kappa_2^2\sqrt{s^2-1}/3\sqrt{3}\kappa_4$. The momenta transported by the solitons are determined by the expression

$$\mathcal{P}_s = \frac{MV}{a^2} \int \rho_0 d\xi = \pm \frac{\pi MV}{a} \sqrt{\frac{\kappa_2}{\kappa_4}}.$$

The plus sign refers to solitons, the minus sign to the antisolitons.

Since the right-hand side of Eq. (5.1) is positive, when seeking consistent solutions of the system of equations (3.2) and (5.1) we must consider only positive values of ρ . In particular, in the first approximation we can substitute in Eq. (3.2) the positive solution (5.3). We then obtain the equation

$$\left(i\hbar \frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m^*} \frac{\partial^2}{\partial \xi^2} + \frac{\Lambda}{\text{ch } \mu \xi} \right) A_1(x, t) = 0, \quad (5.6)$$

where

$$\Lambda = 2\chi\sqrt{2(s^2-1)\kappa_2/\kappa_4}, \quad s^2 > 1. \quad (5.7)$$

The value found for $A_1(x, t)$ enables us to determine the correction ρ_1 to ρ_0 by means of Eq. (5.1). Substituting the value $\rho = \rho_0 + \rho_1$ in Eq. (3.2), we find the value of $A(x, t)$ in the next approximation.

The final term in the round brackets in Eq. (5.6) is the potential well for the external electron. To investigate the solution of this equation, we use the transformation (4.14). Then the wave function $\varphi(\xi)$ of the electron in the coordinate system moving with the soliton satisfies the equation

$$\frac{\hbar^2}{2m^*} \frac{\partial^2 \varphi}{\partial \xi^2} + \left(E + \frac{\Lambda}{\text{ch } \mu \xi} \right) \varphi = 0, \quad (5.8)$$

where

$$E = \hbar\omega - \mathcal{E}_0 - \frac{1}{2}m^*V^2. \quad (5.8a)$$

We use the variational principle to determine the solutions of Eq. (5.8) with least energy:

$$E_0 = \min \frac{1}{a} \int \varphi \left[-\frac{\hbar^2}{2m^*} \frac{\partial^2 \varphi}{\partial \xi^2} - \frac{\Lambda \varphi}{\text{ch } \mu \xi} \right] \varphi d\xi. \quad (5.9)$$

As variational function, we choose the normalized one-parameter function $\varphi(\xi) = \sqrt{\alpha a / \sqrt{2} \text{ch } \mu \xi}$. Then

$$J(\alpha) = \frac{1}{a} \int \varphi(\xi) \left[-\frac{\hbar^2}{2m^*} \frac{\partial^2}{\partial x^2} - \frac{\Lambda}{\text{ch } \mu \xi} \right] \varphi d\xi = -\Lambda + \frac{\hbar^2 \alpha^2}{6m^*} + \frac{\pi \mu^2 \Lambda}{48 \alpha^2},$$

for $\mu^2 < 4\alpha$. From the condition of a minimum of $J(\alpha)$, we find $\alpha_0^2 = \frac{\mu}{2\hbar} \sqrt{\pi m^* \Lambda / 2}$. Therefore, the minimal energy of an electron in the potential well formed by the local deformation of the molecular chain is determined, without allowance for the back reaction of the electron on the soliton, by the expression $E_0 = J(\alpha_0) = -\Lambda + \hbar^2 \alpha_0^2 / 3m^*$. This energy is negative if $\Lambda > \hbar^2 \alpha_0^2 / 3m^*$.

Charge transport by a soliton in quasideimensional system with anharmonicity of the vibrations of fourth degree without allowance for the back reaction of the electron on the soliton was investigated by Zmuidzinis [10] with the aim of explaining the anomalous increase in the conductivity of TTF-TCNQ crystals at low temperatures.

To elucidate the possibility of using the zeroth approximation, we calculate the first correction to the solution $\rho_0(\xi)$ of Eq. (5.1). We substitute in this equation the value

$$\rho = \rho_0 + \rho_1. \quad (5.10)$$

We assume that

$$\rho_1(\xi) \ll \rho_0(\xi), \quad (5.11)$$

and then, substituting (5.10) in Eq. (5.1), we obtain the linearized equation

$$\frac{a^2}{12} \frac{\partial^2 \rho_1}{\partial \xi^2} - (s^2 - 1) \rho_1 + \frac{3\kappa_4}{\kappa_2} \rho_0^2 \rho_1 = \frac{2\chi}{\kappa_2} |A_1|^2.$$

Substituting in this equation the value (5.3) and $|A_1|^2 = \varphi^2(\xi)$, we find the equation

$$\frac{a^2}{12} \frac{\partial^2 \rho_1}{\partial \xi^2} - (s^2 - 1) \rho_1 + \frac{\beta}{ch^2 \mu \xi} \rho_1 = \frac{C}{ch^2 \alpha_0 \xi}, \quad (5.12)$$

where

$$B = 6(s^2 - 1), \quad C = a\chi\alpha_0/\kappa_2. \quad (5.13)$$

Setting $\rho_1 = \text{const } t / \cosh \mu \xi$, we obtain

$$\rho_1 = \frac{3\chi\alpha_0}{a\kappa_2\mu^2} \frac{ch^2 \mu \xi}{ch^2 \alpha_0 \xi}. \quad (5.14)$$

The inequality (5.11), which ensures the validity of perturbation theory to solve Eq. (5.12), now takes the form

$$\frac{3\chi\alpha_0 \sqrt{6\kappa_4}}{a^2 \mu^3 \sqrt{\kappa_2^3}} \frac{ch^3 \mu \xi}{ch^2 \alpha_0 \xi} \ll 1. \quad (5.15)$$

This inequality can be satisfied for all values of ξ under the conditions

$$3\mu < 2\alpha_0 \quad (5.16)$$

and

$$\frac{3\chi\alpha_0}{a^2 \mu^3} \sqrt{\frac{6\kappa_4}{\kappa_2^3}} \ll 1. \quad (5.17)$$

The inequalities (5.13), (5.16), and (5.17) impose fairly stringent conditions on the parameters of a molecular chain with quadratic anharmonicity capable of transporting electrons by solitons.

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